

CIS 5200 Recitation 1: Mathematical Foundations and Learning Basics

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Linear Algebra Foundations

Machine-learning models are built from vectors and matrices. A data point is often represented by a vector, a dataset by a matrix, and a model by a sequence of transformations. Linear algebra gives us a compact language for describing those transformations and understanding when information is preserved or lost.

Matrix products, trace, determinant, transpose, and inverse

If $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$, then $AB \in \mathbb{R}^{m \times p}$. The inner dimensions must agree. Multiplication represents composition: in

$$ABx,$$

the vector x is transformed by B first and by A second. This right-to-left order is why matrix multiplication usually does not commute:

$$AB \neq BA.$$

For example, let

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}, \quad BA = \begin{bmatrix} 2 & 2 \\ 0 & 1 \end{bmatrix}.$$

The first matrix shears and the second stretches. Stretching before shearing is not the same operation as shearing before stretching.

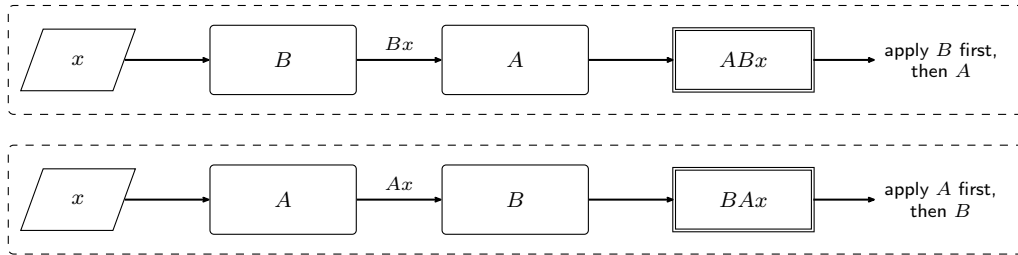


Figure 1: Matrix multiplication composes transformations from right to left. Changing the order changes the intermediate vector and usually changes the final result.

Trace

For a square matrix, the trace is the sum of the diagonal entries:

$$\text{tr}(A) = \sum_i A_{ii}.$$

Trace is linear:

$$\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B), \quad \text{tr}(cA) = c \text{tr}(A).$$

It is also cyclic:

$$\text{tr}(AB) = \text{tr}(BA),$$

and, for three compatible matrices,

$$\text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB).$$

Cyclic does not mean that factors may be reordered arbitrarily. For example, $\text{tr}(ABC)$ and $\text{tr}(ACB)$ can differ. A safe mental model is that the first factor may be moved to the end, as though the expression were written around a circle.

Determinant

For a square matrix, the determinant describes how the associated linear transformation scales signed volume. It satisfies

$$\det(AB) = \det(A) \det(B).$$

If $\det(A) = 0$, the transformation collapses at least one direction and therefore cannot be inverted. Unlike trace, determinant is not additive:

$$\det(A + B) \neq \det(A) + \det(B)$$

in general.

Transpose and inverse

Transpose and inverse both reverse the order of a product:

$$(AB)^\top = B^\top A^\top, \quad (AB)^{-1} = B^{-1} A^{-1}.$$

The inverse identity is intuitive from composition. If AB applies B and then A , undoing the full operation requires undoing A first and then B . The inverse formula requires both matrices to be square and invertible. Trace and determinant similarly require square matrices, while the product itself requires compatible dimensions.

Concept question: matrix identities

Let A , B , and C be square matrices of the same size. Which identities always hold? More than one may be correct.

- A. $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$
- B. $\det(A + B) = \det(A) + \det(B)$
- C. $(ABC)^\top = C^\top B^\top A^\top$
- D. $\text{tr}(ABC) = \text{tr}(CAB)$

Answer: A, C, and D. Trace is linear, transpose reverses factor order, and trace permits cyclic rotations. Choice B is false because determinants multiply over matrix products but do not distribute over sums.

Eigenvalues and eigenvectors

An eigenvector is a nonzero direction that a matrix does not rotate away from itself. If

$$Av = \mu v, \quad v \neq 0,$$

then v is an eigenvector and μ is its eigenvalue. The vector may be stretched, compressed, or reversed, but it stays on the same line.

Eigenvectors are useful because complicated matrix expressions become scalar expressions along eigenvector directions. If p is a polynomial, then

$$Av = \mu v \implies p(A)v = p(\mu)v.$$

For instance,

$$(3A + I)v = (3\mu + 1)v, \quad A^2v = \mu^2v.$$

The eigenvalues also summarize two important matrix quantities:

$$\det(A) = \prod_i \mu_i, \quad \text{tr}(A) = \sum_i \mu_i,$$

where eigenvalues are counted with algebraic multiplicity. These equations are useful for verifying a computed spectrum. Eigenvalues can repeat, and real nonsymmetric matrices can have complex eigenvalues.

Worked example

Suppose a matrix M has eigenvalues 2 and -1 . Without multiplying any matrices, we know that:

$$M^2 \text{ has eigenvalues } 4 \text{ and } 1,$$

$$3M + I \text{ has eigenvalues } 7 \text{ and } -2,$$

and

$$\det(M) = 2(-1) = -2, \quad \text{tr}(M) = 2 + (-1) = 1.$$

Concept question: transforming eigenvalues

Let M be a symmetric 2×2 matrix with eigenvalues 2 and -1 . Which statements are true?

- A. M is positive semidefinite.
- B. M^2 has eigenvalues 4 and 1.
- C. $3M + I$ has eigenvalues 7 and -2 .
- D. $\det(M) = -2$.

Answer: B, C, and D. A symmetric matrix is positive semidefinite only when every eigenvalue is nonnegative. The negative eigenvalue makes A false. The polynomial rule gives B and C, while the determinant is the product of the eigenvalues.

Positive semidefinite matrices

A real symmetric matrix A is positive semidefinite, abbreviated PSD, when

$$x^\top Ax \geq 0 \quad \text{for every } x.$$

For a symmetric matrix, this is equivalent to requiring all eigenvalues to be nonnegative:

$$A \succeq 0 \iff \mu_i \geq 0 \text{ for every } i.$$

The quadratic form $x^\top Ax$ measures curvature or squared size associated with A . Requiring it to be nonnegative says that no direction has negative curvature. Zero is allowed, so a PSD matrix may have flat directions and may be singular. If the inequality is strict for every nonzero x , the matrix is positive definite.

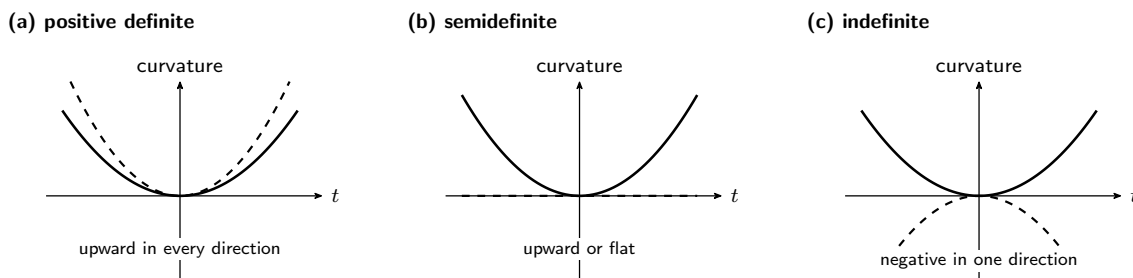


Figure 2: Positive semidefiniteness rules out negative curvature: a positive-definite bowl curves upward in every direction, a PSD surface may contain a flat direction, and a saddle has a direction of negative curvature.

Why $A^\top A$ is always PSD

For any real matrix A and any vector x ,

$$x^\top A^\top A x = (Ax)^\top (Ax) = \|Ax\|_2^2 \geq 0.$$

Therefore,

$$A^\top A \succeq 0.$$

This is one of the most important reusable arguments in linear algebra: rewrite a quadratic form as a squared norm.

PSD matrices are also closed under addition. If $A \succeq 0$ and $B \succeq 0$, then

$$x^\top (A + B)x = x^\top Ax + x^\top Bx \geq 0,$$

so $A + B \succeq 0$.

A positive trace alone does not imply PSD. Trace gives only the sum of the eigenvalues; a positive sum can still hide a negative eigenvalue. For example, $\text{diag}(2, -1)$ has trace 1 but is not PSD.

Concept question: PSD constructions

Which matrices are always positive semidefinite?

- A. $A^\top A$ for any real matrix A .
- B. $A + B$ when A and B are both PSD.
- C. The negative identity matrix $-I$.
- D. Any symmetric matrix with positive trace.

Answer: A and B. Choice A is a squared-norm construction, and PSD quadratic forms add. Choice C has eigenvalues -1 . Choice D is false because a positive sum of eigenvalues does not require each eigenvalue to be nonnegative.

Rank and invertibility

The rank of a matrix is the number of independent directions it preserves. For $A \in \mathbb{R}^{m \times n}$,

$$\text{rank}(A) = \dim(\text{rowspan}(A)) = \dim(\text{colspan}(A)) \leq \min(m, n).$$

Row rank and column rank are always equal, even though the row and column spaces live in different ambient spaces. Computationally, row-reduce the matrix and count the pivot positions or nonzero rows.

For a square $n \times n$ matrix, several statements express the same idea:

$$\text{rank}(A) = n \iff \det(A) \neq 0 \iff A^{-1} \text{ exists} \iff 0 \text{ is not an eigenvalue of } A.$$

If rank is less than n , some nonzero input lies in the null space and is mapped to zero. Two different inputs can then produce the same output, so the transformation cannot be undone uniquely.

Concept question: a rank-two matrix

Let A be a 3×3 matrix with rank 2. Which statements must be true?

- A. A is invertible.
- B. $\det(A) = 0$.
- C. Zero is an eigenvalue of A .
- D. A has exactly two nonzero eigenvalues.

Answer: B and C. Rank two is not full rank, so the matrix is singular, its determinant is zero, and zero is an eigenvalue. Choice D need not hold for a nonsymmetric matrix: rank and the number of nonzero eigenvalues are not generally equal. A nilpotent matrix can have positive rank while every eigenvalue is zero.

Linear-algebra summary

- Products compose transformations from right to left, so order matters.
- Trace is linear and cyclic; determinant is multiplicative.
- Transpose and inverse reverse product order.
- Eigenvectors reduce matrix polynomials to scalar polynomials.
- A symmetric PSD matrix has no direction with a negative quadratic form.
- Rank measures preserved independent directions; full rank is equivalent to invertibility for square matrices.

Probability Foundations

Probability describes uncertainty in observations, labels, and future examples. The central operations are restricting attention after observing evidence, reversing conditional probabilities, and summarizing the center and spread of random variables.

Conditioning and independence

For events A and B with $P(B) > 0$,

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

Conditioning means that B is known to have occurred. Outcomes outside B are discarded, and the probabilities remaining inside B are renormalized so they sum to one. The denominator $P(B)$ performs this renormalization.

Rearranging the definition gives the multiplication rule:

$$P(A \cap B) = P(A | B)P(B).$$

Two events are independent when learning one does not change the probability of the other:

$$A \perp B \iff P(A \cap B) = P(A)P(B) \iff P(A | B) = P(A).$$

Independence must be established mathematically. It is not the same as saying that events sound unrelated. It is also different from mutual exclusivity: if two positive-probability events cannot occur together, observing one makes the other impossible, so they are dependent.

Worked example: two dice

Two fair dice are rolled. Given that at least one die shows a six, what is the probability that both show six?

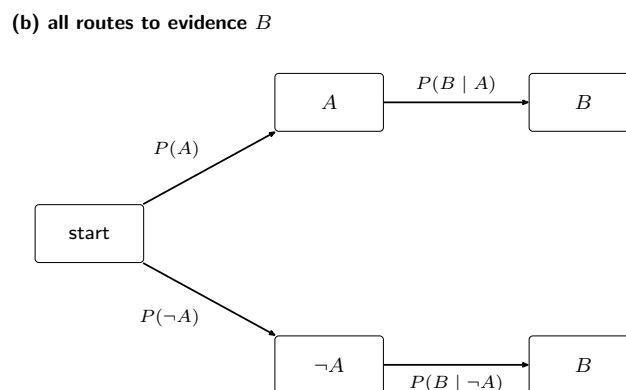
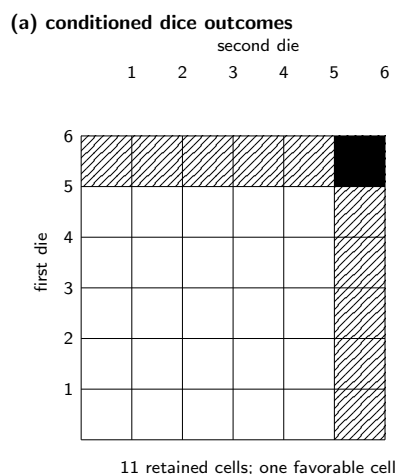


Figure 3: Conditioning keeps only outcomes consistent with the evidence. In the dice example, the original 36 outcomes shrink to 11; in a probability tree, Bayes rule compares the branches that can produce the observed evidence.

Represent outcomes as ordered pairs. Six pairs have the first die equal to six, and six have the second die equal to six. The pair (6,6) is counted in both lists, leaving

$$6 + 6 - 1 = 11$$

possible outcomes after conditioning. Only one is favorable, so

$$P(\text{both six} \mid \text{at least one six}) = \frac{1}{11}.$$

Concept question: conditioned dice

Given that at least one of two fair dice shows a six, what is the probability that both show six?

- A. $1/6$
- B. $1/11$
- C. $1/12$
- D. $1/36$

Answer: B. The conditioned sample space contains eleven equally likely ordered pairs. Choice A would apply if a particular die were known to be six. Choice D ignores the new information.

Bayes rule and total probability

Bayes rule reverses the direction of a conditional probability:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}.$$

It follows directly from writing the same intersection in two ways. Start from

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

and

$$P(B | A) = \frac{P(B \cap A)}{P(A)}.$$

Because $A \cap B = B \cap A$,

$$P(A \cap B) = P(B | A)P(A).$$

Substituting this into the first equation gives Bayes rule. Equivalently,

$$\boxed{P(A \cap B) = P(A | B)P(B) = P(B | A)P(A)}.$$

The terms have useful names:

- $P(A)$ is the **prior**, the probability before observing B .
- $P(B | A)$ is the **likelihood** of the evidence when A is true.
- $P(A | B)$ is the **posterior**, the updated probability after observing B .

A compact memory aid is

$$\text{posterior} \propto \text{likelihood} \times \text{prior}.$$

The denominator accounts for every way the evidence can occur. Splitting the world into A and $\neg A$ gives the law of total probability:

$$P(B) = P(B | A)P(A) + P(B | \neg A)P(\neg A).$$

Worked example: diagnostic testing

Suppose

$$P(D) = 0.01, \quad P(+ | D) = 0.90, \quad P(+ | \neg D) = 0.10.$$

The sensitivity $P(+ | D)$ is not the desired posterior $P(D | +)$. For a population of 10,000 people:

Group	People	Positive results
Diseased	100	90 true positives
Not diseased	9,900	990 false positives
Total	10,000	1,080 positives

Thus,

$$P(D | +) = \frac{90}{1080} \approx 0.0833.$$

The same answer follows from

$$P(D | +) = \frac{0.9(0.01)}{0.9(0.01) + 0.1(0.99)} \approx 0.0833.$$

The positive result is evidence for disease, but the disease is rare. The much larger healthy group produces many false positives even with a modest false-positive rate. This is the base-rate effect.

Concept question: interpreting a positive result

A disease affects 1% of a population. A test has 90% sensitivity and a 10% false-positive rate. A person tests positive. Approximately what is the probability that the person has the disease?

- A. 8%
- B. 45%
- C. 90%
- D. 99%

Answer: A. The numerator is $0.9(0.01) = 0.009$, while the total probability of a positive result is $0.009 + 0.1(0.99) = 0.108$. Their ratio is about 8.3%. Choice C confuses sensitivity with the posterior.

Expectation and variance

Expectation is a probability-weighted average. For a discrete random variable,

$$\mathbb{E}[X] = \sum_x xP(X = x).$$

Values that occur more frequently receive more weight. If the same experiment is repeated many times, the average of the observed values tends toward $\mathbb{E}[X]$. Expectation is therefore the long-run center of a distribution, although it need not be a possible outcome.

For a fair six-sided die,

$$\mathbb{E}[X] = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = 3.5.$$

No individual roll equals 3.5, but the average of many rolls approaches it. Expectation can also be viewed as a balance point: values farther from the center exert more leverage, weighted by their probability.

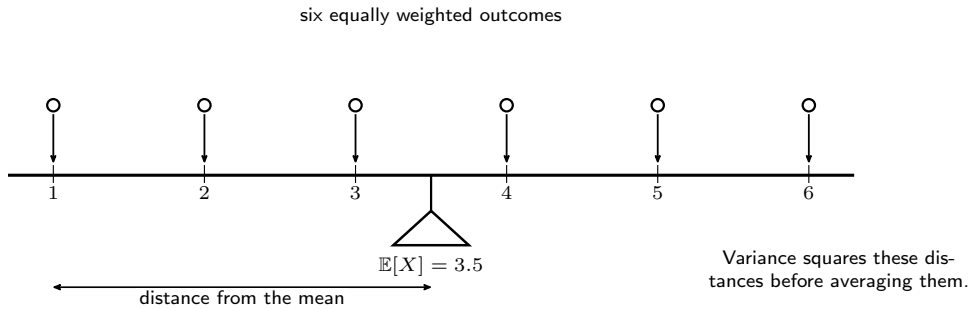


Figure 4: Expectation is the balance point of a probability distribution. Variance averages squared distances from that center, so distant outcomes contribute disproportionately.

Expectation is linear:

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

No independence assumption is required. This remains true even if X and Y are strongly dependent.

Variance measures average squared distance from the mean. If $\mu = \mathbb{E}[X]$, then

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2].$$

Expanding the square gives

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \\ &= \mathbb{E}[X^2] - \mu^2.\end{aligned}$$

Therefore,

$$\boxed{\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2}.$$

The first term is the average squared value. Subtracting the square of the average converts it into the average squared distance from the mean.

Affine transformations obey

$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$

Adding b moves the entire distribution without altering its spread. Multiplying by a multiplies every distance from the mean by $|a|$, so squared distances scale by a^2 .

For two random variables,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y).$$

If X and Y are independent, their covariance is zero, and their variances add. Independence is needed for this simplification, not for linearity of expectation or for the affine variance rule.

Concept question: moments under transformations

Suppose $\mathbb{E}[X] = 2$ and $\text{Var}(X) = 3$. Which statements are true?

- A. $\mathbb{E}[2X + 1] = 5$.
- B. $\text{Var}(2X + 1) = 13$.
- C. $\mathbb{E}[X^2] = 7$.
- D. $\text{Var}(-X) = -3$.

Answer: A and C. Linearity gives $2(2) + 1 = 5$. The affine variance rule gives $2^2(3) = 12$, not 13. Rearranging the variance identity gives $\mathbb{E}[X^2] = 3 + 2^2 = 7$. Finally, $\text{Var}(-X) = (-1)^2 \text{Var}(X) = 3$; variance cannot be negative.

Gaussian random variables

A Gaussian random variable is described by its mean and variance:

$$Z \sim \mathcal{N}(\mu, \sigma^2).$$

The mean locates the center of the bell curve, and the standard deviation σ sets its horizontal scale. In this notation, the second parameter is the variance σ^2 , not the standard deviation.

Gaussians are closed under affine transformations:

$$Z \sim \mathcal{N}(\mu, \sigma^2) \implies aZ + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2).$$

Standardization is the special transformation

$$U = \frac{Z - \mu}{\sigma} \sim \mathcal{N}(0, 1).$$

Subtracting μ recenters the distribution at zero. Dividing by σ changes the horizontal ruler so that one standard deviation becomes one unit. The numerical coordinates change, but relative unusualness does not: a point one standard

deviation above the original mean becomes +1.

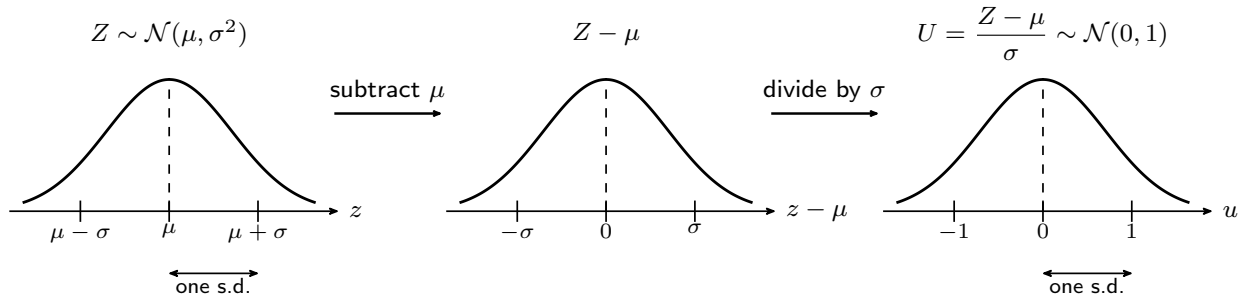


Figure 5: Standardization is recentering and rescaling. Subtracting μ moves the center to zero; dividing by σ changes the horizontal units to standard deviations.

If Z and $\bar{Z}$ are independent Gaussians, then their sum is also Gaussian:

$$Z \sim \mathcal{N}(\mu, \sigma^2), \quad \bar{Z} \sim \mathcal{N}(\bar{\mu}, \bar{\sigma}^2), \quad Z \perp \bar{Z},$$

implies

$$Z + \bar{Z} \sim \mathcal{N}(\mu + \bar{\mu}, \sigma^2 + \bar{\sigma}^2).$$

This equation combines two separate bell-shaped sources; it does not standardize one variable. Their centers add, and their variances add because independence removes the covariance term. Standard deviations do not add directly.

Concept question: Gaussian rules

Suppose

$$Z \sim \mathcal{N}(1, 4), \quad \bar{Z} \sim \mathcal{N}(2, 1), \quad Z \perp \bar{Z}.$$

Which statements are true?

- A. $(Z - 1)/2$ is standard normal.
- B. $3Z$ is Gaussian with mean 3 and variance 36.
- C. $Z + \bar{Z}$ is Gaussian with mean 3 and variance 3.
- D. $\mathbb{E}[Z^2] = 5$.

Answer: A, B, and D. The standard deviation of Z is 2, so A standardizes correctly. Multiplying by 3 multiplies the variance by 9, giving B. For C, the variances add to $4 + 1 = 5$, not 3. The second-moment identity gives $\mathbb{E}[Z^2] = 4 + 1^2 = 5$.

Probability summary

- Conditioning restricts the sample space and renormalizes probability.
- Independence means that conditioning does not alter the other event's probability.
- Bayes rule combines likelihood and prior, then normalizes over all explanations of the evidence.
- Expectation is a weighted average and is always linear.
- Variance measures squared spread; independent variances add.
- Standardization recenters and rescales one Gaussian, while the independent-sum rule combines two Gaussians.

Convexity and Geometry

Optimization asks us to choose parameters that minimize a loss. Convexity tells us when the landscape has no deceptive local minima, while norms specify how vector size and distance are measured.

Convex functions

A function f is convex when, for any x , y , and $\alpha \in [0, 1]$,

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y).$$

The input

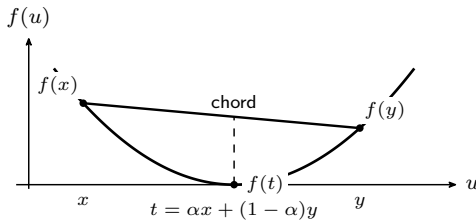
$$t = \alpha x + (1 - \alpha)y$$

is a point between x and y . At that horizontal location, $f(t)$ is the height of the graph. The quantity

$$\alpha f(x) + (1 - \alpha)f(y)$$

is the height of the straight chord connecting $(x, f(x))$ and $(y, f(y))$. Convexity says that the graph stays at or below that chord. A stretched string between two points on a bowl lies above the bowl.

(a)



(b)

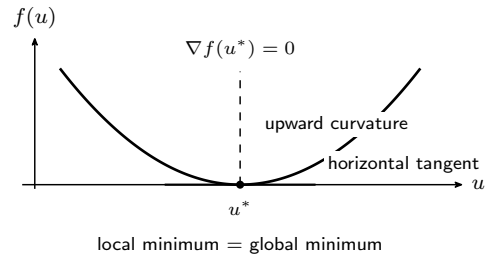


Figure 6: Convexity has two visual consequences: every chord lies above the graph, and a stationary local minimum of a differentiable convex function is global.

For a twice-differentiable function on a convex domain,

$$f \text{ is convex} \iff \nabla^2 f(x) \succeq 0 \quad \text{for every } x.$$

The Hessian $\nabla^2 f(x)$ is the matrix of second derivatives. For a direction v ,

$$v^\top \nabla^2 f(x) v$$

is the second derivative of f along that direction. A PSD Hessian therefore means that the function curves upward or remains flat in every direction. In one dimension, the condition reduces to $f''(x) \geq 0$.

Worked example: bowl versus saddle

For

$$f(x_1, x_2) = x_1^2 + 3x_2^2, \quad \nabla^2 f = \begin{bmatrix} 2 & 0 \\ 0 & 6 \end{bmatrix},$$

we have

$$v^\top \nabla^2 f v = 2v_1^2 + 6v_2^2 \geq 0.$$

The surface curves upward in every direction and is convex. By contrast,

$$g(x_1, x_2) = x_1^2 - x_2^2, \quad \nabla^2 g = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}.$$

Taking $v = (0, 1)$ gives $v^\top \nabla^2 g v = -2$. The function curves downward in that direction, producing a saddle.

The optimization payoff is decisive: every local minimum of a convex function is global. If a differentiable convex function satisfies $\nabla f(x^*) = 0$, then x^* is a global minimizer. Convex functions are closed under addition and nonnegative scaling.

The chord definition is more fundamental than the Hessian condition because it also handles nondifferentiable convex functions such as $|x|$.

Concept question: global convexity

Which functions are convex on all of $\mathbb{R}$?

- A. e^x
- B. $|x|$
- C. x^3
- D. $\sin x$

Answer: A and B. The exponential has second derivative $e^x > 0$. Absolute value satisfies the chord definition despite being nondifferentiable at zero. The curvature of x^3 changes sign, while the curvature of sine oscillates, so neither is globally convex.

Vector norms

A norm assigns a nonnegative size to a vector. The p -norm family is

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p},$$

with the limiting case

$$\|x\|_\infty = \max_i |x_i|.$$

Every norm satisfies three structural properties:

1. **Definiteness:** $\|x\| \geq 0$, with equality only when $x = 0$.
2. **Absolute homogeneity:** $\|cx\| = |c|\|x\|$.
3. **Triangle inequality:** $\|x + y\| \leq \|x\| + \|y\|$.

The quantity

$$\|x\|_0 = \#\{i : x_i \neq 0\}$$

counts nonzero coordinates but is not a true norm. For nonzero c , scaling the vector usually leaves the count unchanged, so $\|cx\|_0 = \|x\|_0$ rather than $|c|\|x\|_0$.

Geometry of the unit balls

The unit ball of a norm is the set $\{x : \|x\| \leq 1\}$. In two dimensions:

- The ℓ_1 unit ball is a diamond because $|x_1| + |x_2| \leq 1$ creates a fixed coordinate budget. Increasing one coordinate leaves less magnitude for the other.

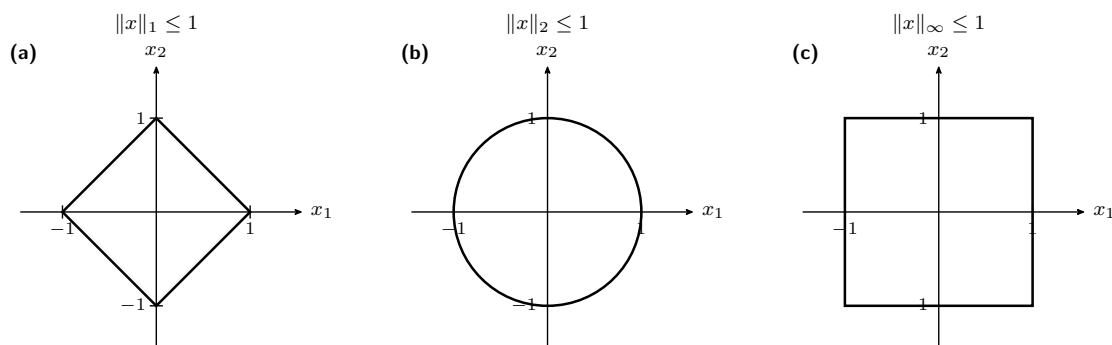


Figure 7: The 1-, 2-, and infinity-norm unit balls are a diamond, circle, and square. Moving along each boundary changes the vector but keeps its norm equal to one. The shape tells us how the norm allows the coordinates to trade off while preserving that size.

- The ℓ_2 unit ball is a circle because $x_1^2 + x_2^2 \leq 1$ is ordinary Euclidean distance.
- The ℓ_∞ unit ball is a square because each coordinate independently satisfies $|x_i| \leq 1$.

Moving outward from the origin crosses successively larger level sets, so the norm increases. Moving along one boundary changes the vector's direction and coordinate allocation while keeping its norm fixed.

The nesting

$$B_1 \subseteq B_2 \subseteq B_\infty$$

corresponds to

$$\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1.$$

In finite dimensions all p -norms are equivalent up to dimension-dependent constants. When examining a proposed norm inequality, two particularly informative vectors are a standard basis vector e_i and the all-ones vector. The basis vector often makes all common norms equal to one; the all-ones vector exposes how the constant depends on dimension.

Concept question: the all-ones vector

Let $x = (1, 1, \dots, 1) \in \mathbb{R}^n$. Which statements are true?

- A. $\|x\|_1 = n$.
- B. $\|x\|_2 = \sqrt{n}$.
- C. $\|x\|_\infty = 1$.
- D. $\|x\|_0 = 1$.

Answer: A, B, and C. Summing magnitudes gives n , summing squares and taking a square root gives $\sqrt{n}$, and the largest coordinate magnitude is one. Choice D is false because all n coordinates are nonzero, so $\|x\|_0 = n$.

Foundations of Machine Learning

Machine learning uses data to construct predictors that perform well on examples not seen during training. The central difficulty is not merely fitting observed data; it is controlling the difference between performance on the sample and performance on the underlying population.

Three learning paradigms

Supervised learning

Supervised learning observes input-label pairs (x, y) and learns to predict labels for new inputs. Classification predicts discrete labels, such as spam versus not spam. Regression predicts numerical values, such as tomorrow's electricity

demand.

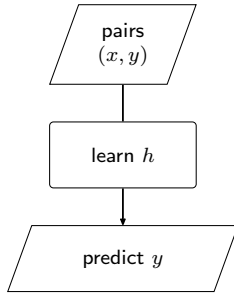
Unsupervised learning

Unsupervised learning observes inputs without labels and looks for structure. Clustering customers by purchasing behavior is a canonical example. The desired structure is not supplied directly as a target label.

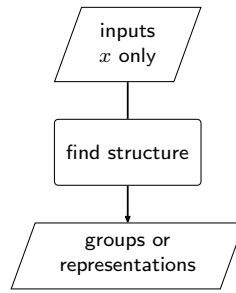
Reinforcement learning

In reinforcement learning, an agent repeatedly observes a state, chooses an action, and receives a reward. Its decisions influence later states, so the problem involves sequential consequences rather than independent labeled examples.

(a) supervised



(b) unsupervised



(c) reinforcement

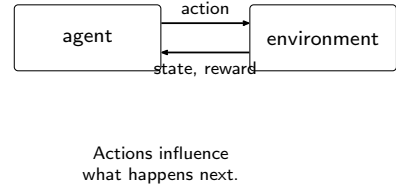


Figure 8: The learning paradigms differ in the information supplied: supervised learning provides targets, unsupervised learning provides only observations, and reinforcement learning provides rewards produced by sequential interaction.

Concept question: identifying the paradigm

Classify each scenario as supervised, unsupervised, or reinforcement learning.

- A. Predict tomorrow's electricity demand from past demand records.
- B. Group customers by purchasing behavior with no labels.
- C. Train a robot to walk using reward for forward motion.
- D. Flag email as spam using user-labeled messages.

Answer: A and D are supervised, B is unsupervised, and C is reinforcement learning. The essential clues are whether explicit targets are supplied and whether actions affect an evolving environment.

The supervised-learning setup

A training dataset contains n examples:

$$D = ((x_1, y_1), \dots, (x_n, y_n)), \quad x_i \in \mathcal{X} \subseteq \mathbb{R}^d, \quad y_i \in \mathcal{Y}.$$

The predictor, or hypothesis, is a function

$$h : \mathcal{X} \rightarrow \mathcal{Y}, \quad \hat{y} = h(x).$$

A loss function $\ell(\hat{y}, y)$ assigns a cost to a prediction. The training error, also called empirical risk, is the average loss on the observed dataset:

$$\hat{R}(h) = \frac{1}{n} \sum_{i=1}^n \ell(h(x_i), y_i).$$

The true risk is the expected loss on a fresh example from the data-generating distribution P :

$$R(h) = \mathbb{E}_{(x,y) \sim P}[\ell(h(x), y)].$$

This is the quantity we ultimately care about, but P is unknown. A held-out test set provides an estimate using examples that were not used to fit the model.

The i.i.d. assumption

Standard learning theory often assumes the examples are independent and identically distributed. Independent means that one sampled example does not alter another; identically distributed means that training and future examples come from the same distribution. This is an assumption about how the data were generated, not a fact guaranteed by the dataset.

When deployment data differ systematically from training data, a test set drawn in the old way can give an overly optimistic estimate. Generalization claims must therefore be interpreted relative to the assumed data distribution.

Generalization and memorization

The generalization gap is

$$R(h) - \hat{R}(h).$$

Equivalently,

$$R(h) = \hat{R}(h) + (R(h) - \hat{R}(h)).$$

True risk equals training error plus the generalization gap. A model can therefore achieve tiny training error and still have large true risk.

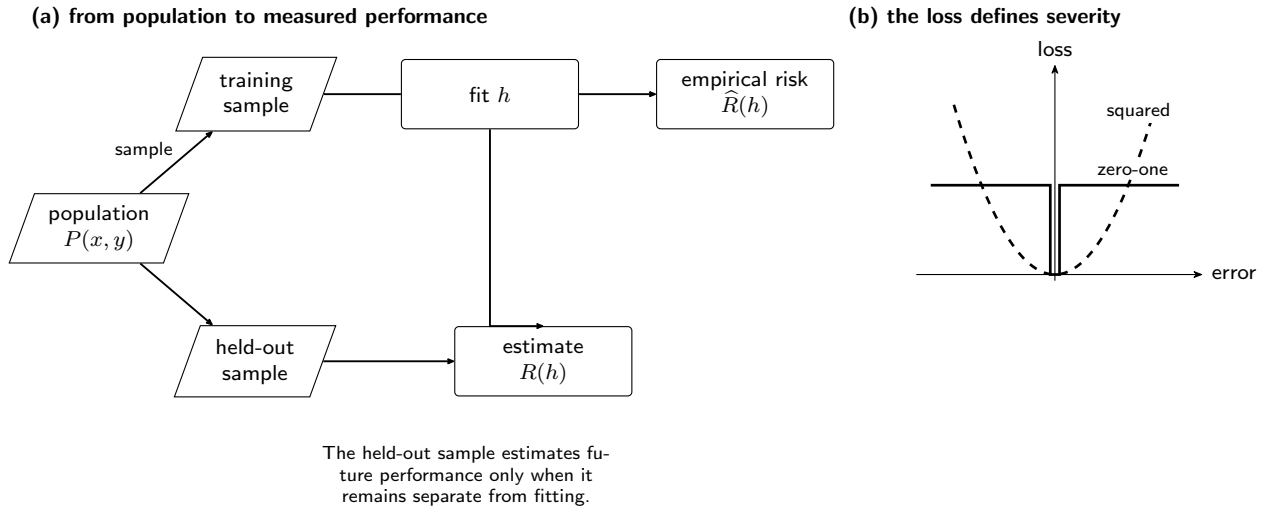


Figure 9: Training minimizes empirical risk on observed examples, but deployment depends on true risk under the population distribution. A held-out test sample estimates the gap only when it represents the same population and remains separate from training.

Worked example: the memorizer

Consider a classifier that stores every training pair. When given an input it has seen, it returns the stored label; for every unseen input, it returns $+1$.

If all training features are distinct, this classifier has zero training error. Yet it has learned no general rule. On a large or continuous input space, a fresh point is almost certainly not an exact training point, so the classifier predicts $+1$ almost everywhere. If many true labels are -1 , the true risk can be large.

The example separates **fitting** from **learning**. Zero training error describes performance on remembered examples; it does not by itself provide evidence about future examples.

Concept question: the memorizing classifier

Which statements about the memorizer must or can be true under the setup above?

- A. Its training error is zero.
- B. Zero training error implies low true risk.
- C. Its generalization gap can be large.
- D. It predicts +1 on almost every new instance in a large or continuous domain.

Answer: A, C, and D. The stored labels guarantee A. Choice B is precisely the mistaken conclusion exposed by the example. The behavior on unseen inputs makes C possible and yields D under the stated domain assumption.

Choosing a loss function

The loss function defines what counts as a serious mistake. It must match the output type and the goal of the task.

For classification, zero-one loss is

$$\ell_{0/1}(\hat{y}, y) = \mathbf{1}\{\hat{y} \neq y\}.$$

It records whether the predicted class is correct, not how close a numerical prediction is. For regression, squared loss is

$$\ell_{\text{sq}}(\hat{y}, y) = (\hat{y} - y)^2.$$

Squared loss distinguishes small numerical misses from large ones and penalizes large errors especially strongly.

Worked example: house-price predictions

Suppose a house sells for \$510,000. Model A predicts \$509,000, while Model B predicts \$3.50.

Under exact-match zero-one loss, both predictions are simply wrong and receive loss one. The loss throws away the crucial fact that Model A is extremely close. Under squared loss,

$$(509,000 - 510,000)^2 = 1,000^2,$$

while Model B's squared error is approximately $509,996.5^2$, making Model A overwhelmingly better.

Concept question: matching loss to task

Which statements are true in the house-price example?

- A. Under zero-one loss, both predictions incur the same loss.
- B. Under squared loss, Model A is far better than Model B.
- C. Zero-one loss is a natural choice for regression.
- D. Squared loss treats a \$1,000 miss and a roughly \$510,000 miss equally.

Answer: A and B. Zero-one loss records only exact equality and is therefore uninformative for most regression problems. Squared loss increases rapidly with error magnitude, so C and D are false.

The Bayes-optimal classifier

For binary labels $y \in \{-1, +1\}$, define

$$\eta(x) = P(y = +1 \mid x).$$

The quantity $\eta(x)$ belongs to the data distribution. It is not defined by a particular fitted model. Under zero-one loss, the best possible classifier predicts the more likely label at each x :

$$h^*(x) = \begin{cases} +1, & \eta(x) > \frac{1}{2}, \\ -1, & \text{otherwise.} \end{cases}$$

Why is this optimal? At a fixed x , predicting $+1$ is wrong with probability $1 - \eta(x)$, while predicting -1 is wrong with probability $\eta(x)$. Choose the smaller of those two error probabilities.

The resulting Bayes risk is

$$R^* = \min_h R(h) = \mathbb{E}_x [\min(\eta(x), 1 - \eta(x))].$$

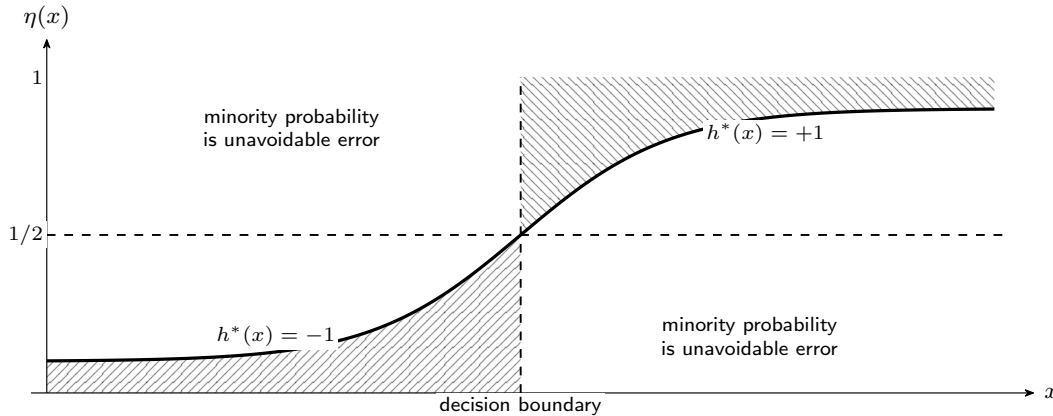


Figure 10: The Bayes classifier predicts the locally more probable class. Where $\eta(x) > 1/2$ it predicts $+1$; the remaining minority probability is irreducible error even for the optimal rule.

Worked example: unavoidable uncertainty

If $\eta(x) = 0.7$ at a particular input, then $+1$ occurs 70% of the time and -1 occurs 30% of the time. The Bayes classifier predicts $+1$, but it is still wrong 30% of the time at that input.

Bayes error is zero only when labels are deterministic given x , meaning $\eta(x)$ equals zero or one almost everywhere. If identical-looking inputs can genuinely receive different labels, some error is irreducible. Learning aims to approach R^* ; no classifier can systematically beat it under the same distribution and loss.

Concept question: applying the Bayes rule

At a particular input x_0 , suppose $\eta(x_0) = 0.7$. Which statements are true?

- A. The Bayes classifier predicts $+1$.
- B. Its conditional error at x_0 is 0.3.
- C. A perfect classifier must achieve zero error at x_0 .
- D. The value 0.7 is a property of the conditional data distribution.

Answer: A, B, and D. The majority label is $+1$, and its minority probability is 0.3. Choice C is false because label uncertainty creates irreducible error.

Machine-learning summary

- Supervised learning uses labeled examples; unsupervised learning searches unlabeled data for structure; reinforcement learning uses sequential rewards.
- Empirical risk is average training loss, while true risk is expected loss on a fresh population example.
- A test set estimates future performance only when it is representative and kept separate from model fitting.
- Zero training error does not imply low true risk.
- The loss function determines which errors matter.

- The Bayes classifier chooses the more likely label at each input and exposes the irreducible error of the data distribution.

Compact Reference

Idea	Core statement	Interpretation
Cyclic trace	$\text{tr}(ABC) = \text{tr}(BCA)$	Rotate factors; do not arbitrarily reorder them.
Product inverse	$(AB)^{-1} = B^{-1}A^{-1}$	Undo transformations in reverse order.
Eigenvalue polynomial	$p(A)v = p(\mu)v$	Matrix algebra becomes scalar algebra along an eigenvector.
PSD	$x^\top Ax \geq 0$	No direction has negative quadratic curvature.
Full rank	$\text{rank}(A) = n \Leftrightarrow A^{-1}$ exists	No input direction is collapsed.
Conditioning	$P(A \mid B) = P(A \cap B)/P(B)$	Restrict to B and renormalize.
Bayes rule	$P(A \mid B) = P(B \mid A)P(A)/P(B)$	Reverse the conditioning direction using the prior.
Expectation	$\mathbb{E}[X] = \sum_x xP(X = x)$	Probability-weighted average.
Variance	$\mathbb{E}[X^2] - \mathbb{E}[X]^2$	Average squared spread around the mean.
Gaussian affine rule	$aZ + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$	Shifts and rescalings preserve Gaussian form.
Convexity	$f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha)f(y)$	Every chord lies above the graph.
Norm	Definiteness, homogeneity, triangle inequality	A consistent notion of vector size.
Empirical risk	$\widehat{R}(h) = n^{-1} \sum_i \ell(h(x_i), y_i)$	Average loss on observed data.
True risk	$R(h) = \mathbb{E}[\ell(h(x), y)]$	Expected loss on a fresh example.
Bayes classifier	Predict +1 when $\eta(x) > 1/2$	Choose the locally more probable label.